

Supplementary Appendix for Public-Private Cooperation in the Port of Rotterdam: An Agent-Based Model of Cooperative Efforts Against Cocaine Trafficking

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1 Acknowledgements

This research is part of the project Focusing on the Right Things in the Port of Rotterdam (with project number KICH1.VE03.21.008) of the research programme KIC-MISSION 2021 which is (partly) financed by the Dutch Research Council (NWO).

A Supplementary Appendix

B Model Architecture

PPC effectiveness is not a directly observed variable but a composite performance index that summarizes whether the conditions required for effective Port Public–Private Cooperation (PPC) are satisfied. It is recalculated at every simulation step and constrained to the interval $[0, 1]$, where 0 represents the lowest attainable modeled effectiveness and 1 represents the highest attainable modeled effectiveness under the implemented rules.

Table 1: Relational Parameter Cross-Impact Matrix

Influencing Parameter	competition	cooperation	trust	pressure	enforcement
competition	x	0.5	0.2	0.2	x
cooperation	0.5	x	0.8	0.5	x
trust	0.8	0.8	x	0.2	x
pressure	0.2	0.8	x	x	0.8
enforcement	x	0.8	x	0.5	x

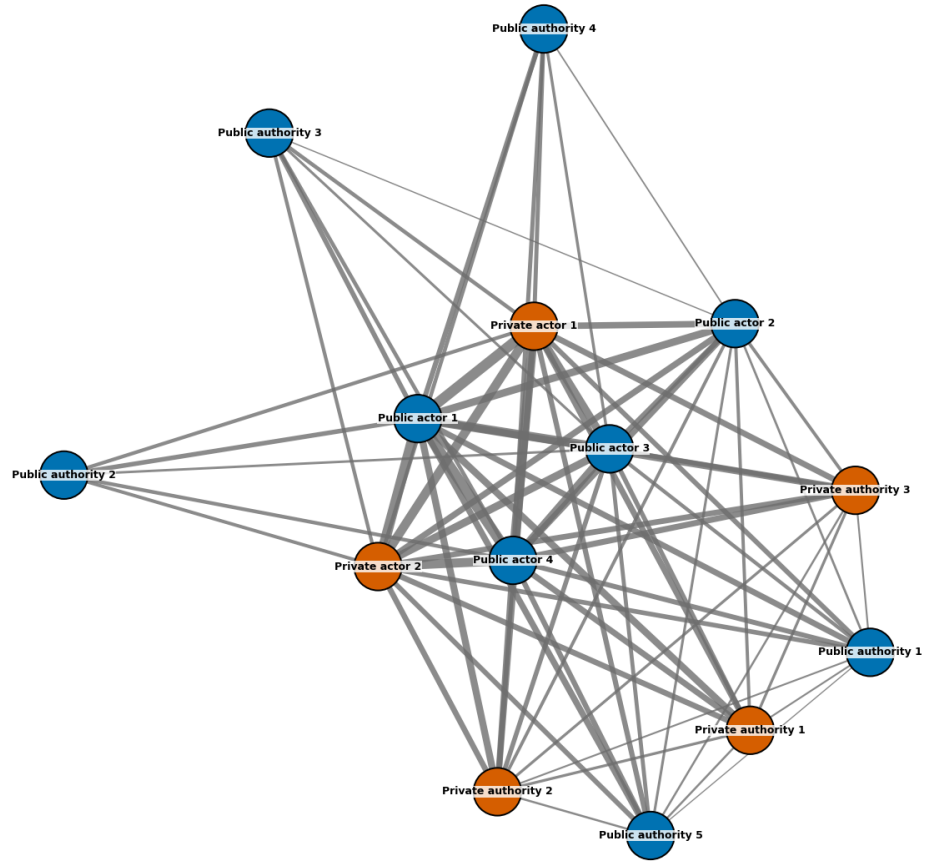


Fig.1: Baseline actor network generated through endogenous tie formation. Nodes represent actors, with blue nodes indicating public actors and orange nodes indicating private actors. Edges represent simulated collaborative ties derived from actor skill profiles, where thicker edges indicate stronger tie strength. The network contains 14 actors and 72 ties. The network density is 0.79, with an average clustering of 0.8 and an average degree of 10.3. The average tie strength is 0.16, and the most central actor is Public Actor 1.

Pressure and Enforcement

B.1 OAT

B.2 PPC effectiveness

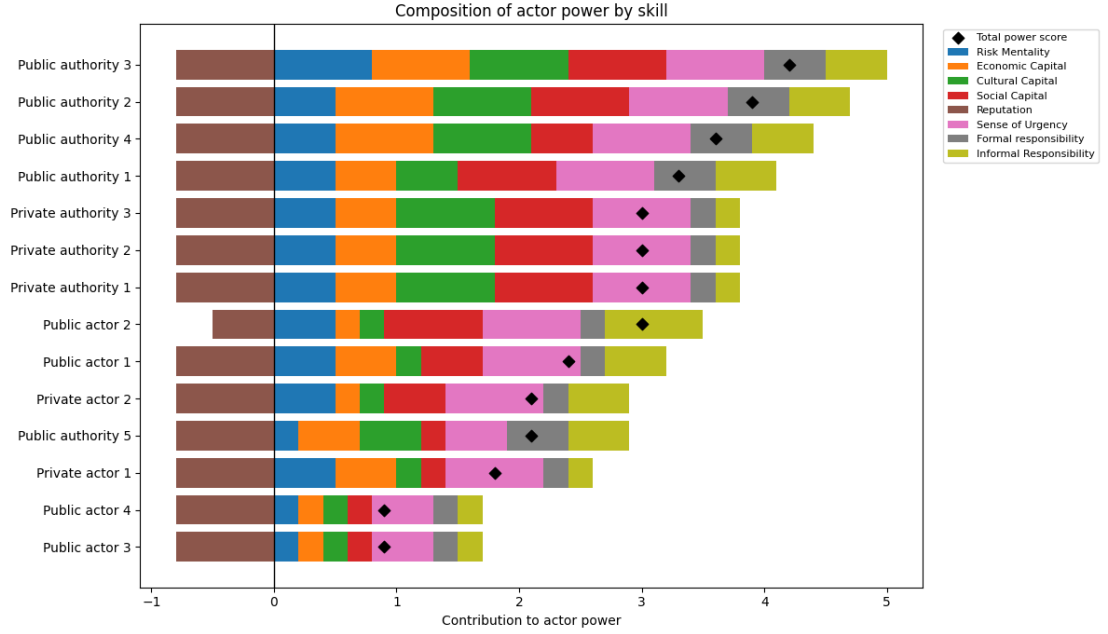


Fig. 2: Actor skills and institutional characteristics used in the ABM initialization. Values represent normalized skill levels between 0 and 1. Skills may change over time through model dynamics. Reputation accounts negatively for the cumulative actor skills.

Table 2: Relational parameter-to-actor skill influence matrix.

Parameter	Risk	Punish.	Enforce.	Press.	Econ.	Cult.	Social	Reput.	Urg.	Formal	Informal
Competition	0.2	0.8	0.5	0.8	0.2	0.2	0.2	0.8	0.5	x	x
Cooperation	0.8	0.2	0.2	0.5	0.2	0.5	0.8	0.2	0.5	0.8	0.5
Trust	0.8	0.2	0.5	0.5	0.5	0.5	0.5	0.2	x	0.5	0.5
Pressure	0.5	0.8	0.2	0.8	0.8	0.5	0.8	0.8	0.8	0.8	0.8
Enforcement	0.8	0.8	0.8	0.5	0.2	x	x	0.5	0.5	0.5	0.2

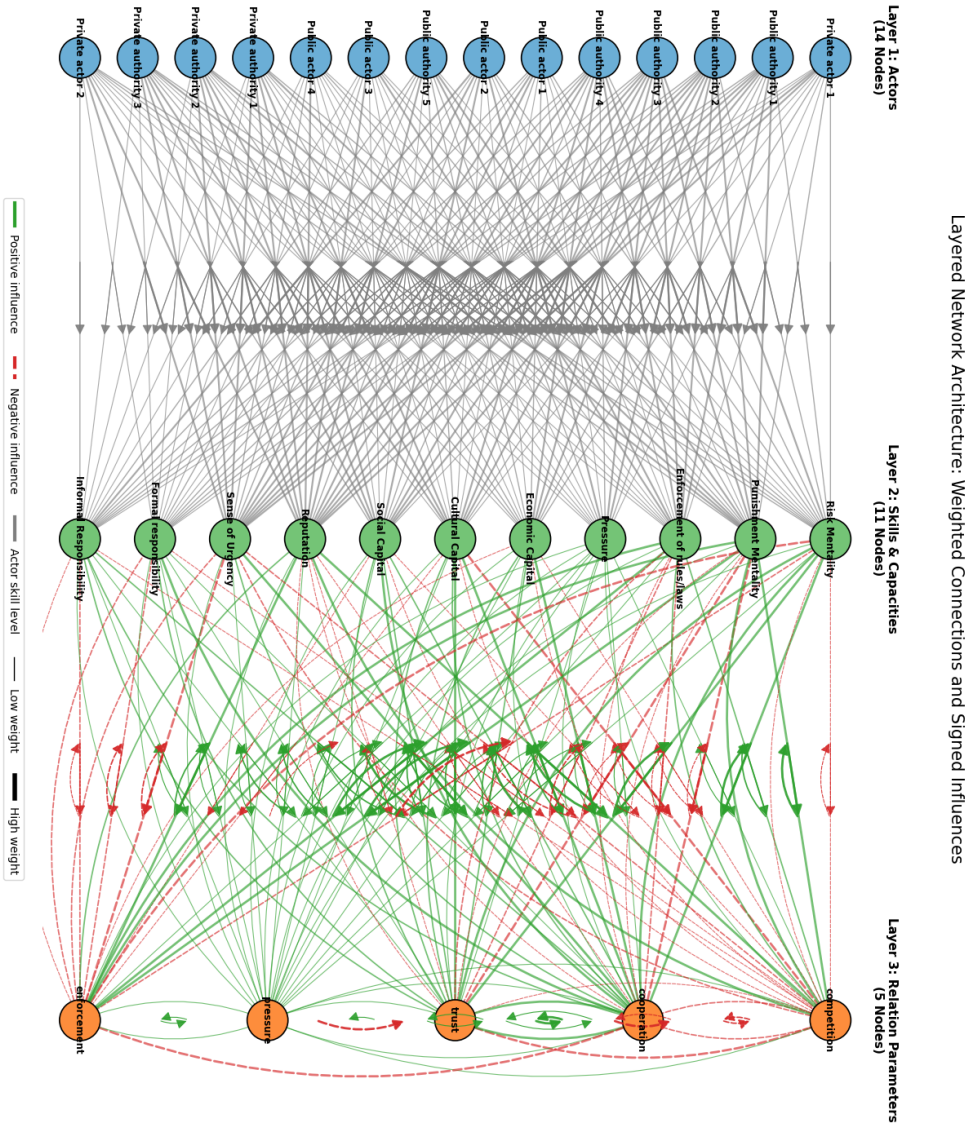


Fig. 3: Conceptual architecture and interconnections of the multi-layer model. Formal institutional mechanisms (e.g., enforcement and pressure) establish the governance baseline, directly influencing edge weights and interaction probabilities within the relational layer (cooperation, trust, and competition). Feedback and propagation dynamics across the relational network subsequently shape aggregate PPC effectiveness over time. Arrows indicate directional information flow, parameter updates, and system feedback loops.

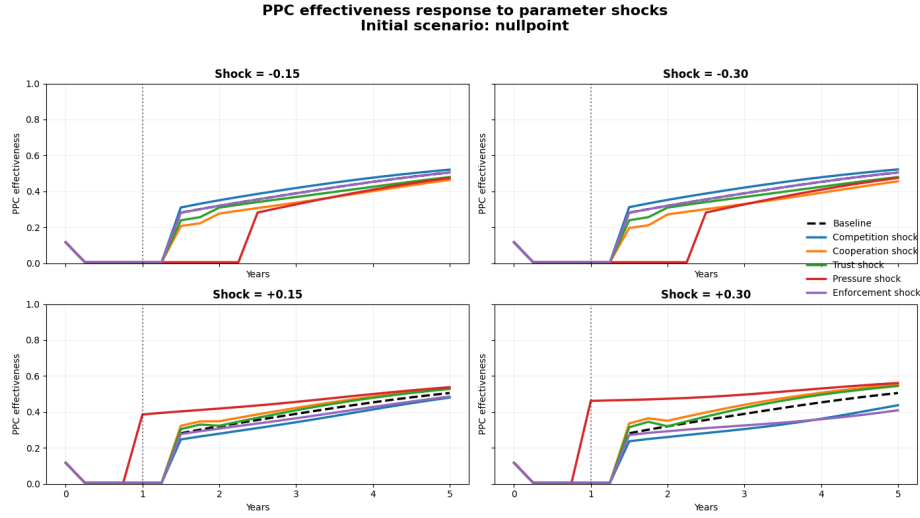


Fig. 4: PPC effectiveness following single-parameter shock interventions under the nullpoint initialization scenario. All relational parameters are initialized at zero, representing the absence of pre-existing relationships. Each line represents the PPC effectiveness trajectory over the five-year simulation period following a single positive or negative parameter shock.

Table 3: ABM and CLD Implementation Traceability Log

Rule / System Component	Type	Influenced Elements	Mapping Assumptions	Logic / Assumptions
Parameter Matrix	relation parameters	Cross-parameter loops	Direct	coefficient transfer into <code>param_to_param</code> structures.
Skill Influence matrix	relation parameters	Actor skill values	Bounded	matrix operations constrained to $[0, 1]$.
Pressure Sweet Spot	rules	Global network cohesion	Evaluation	of total rule compliance through parametric shock sweeps.
Time Discretization	rules	Model step up-dates	Verbatim	definition where 1 step = 3 months (Quarterly increments).
Mentality Disparity	rules	Risk vs. Punishment	Condition checked	via a boolean evaluator inside the success loop.
Capital Sufficiency	rules	Asset availability	Assumed numeric floor	(> 0.1) required to satisfy network health.

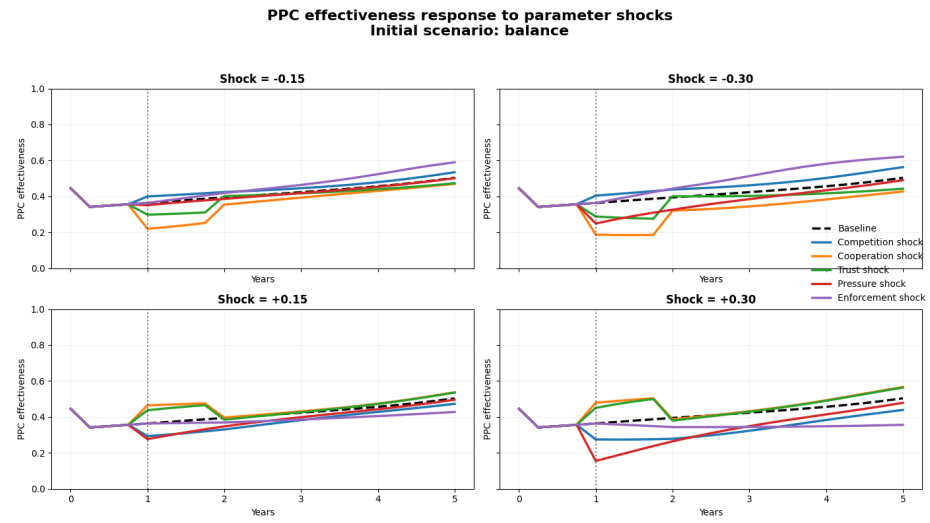


Fig. 5: PPC effectiveness following single-parameter shock interventions under the balanced initialization scenario. All relational parameters are initialized at a neutral value of 0.5, allowing the isolated propagation of shocks through the network from an equilibrium state. Each line represents the PPC effectiveness trajectory over the five-year simulation period following a single positive or negative parameter shock.

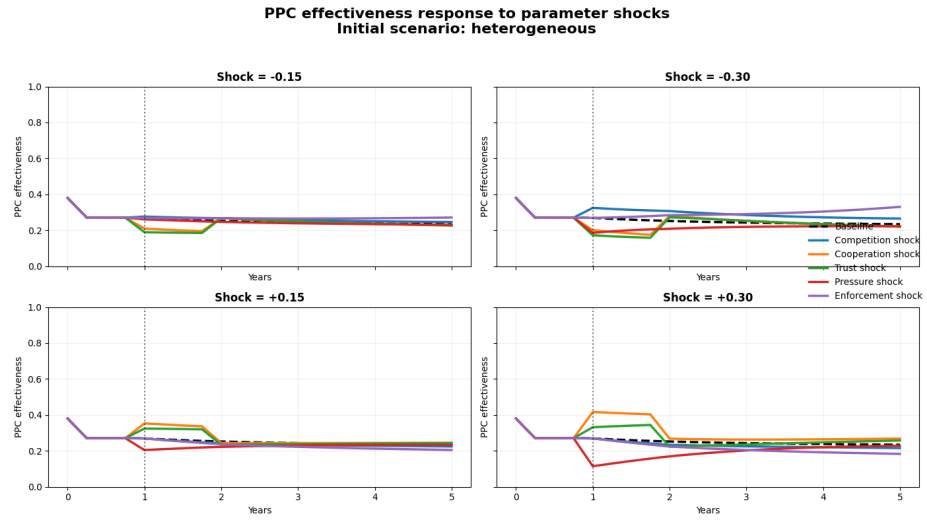


Fig. 6: PPC effectiveness following single-parameter shock interventions under the heterogeneous initialization scenario. Parameter values are derived from expert knowledge, representing a more realistic network configuration with existing relational differences. Each line represents the PPC effectiveness trajectory over the five-year simulation period following a single positive or negative parameter shock.

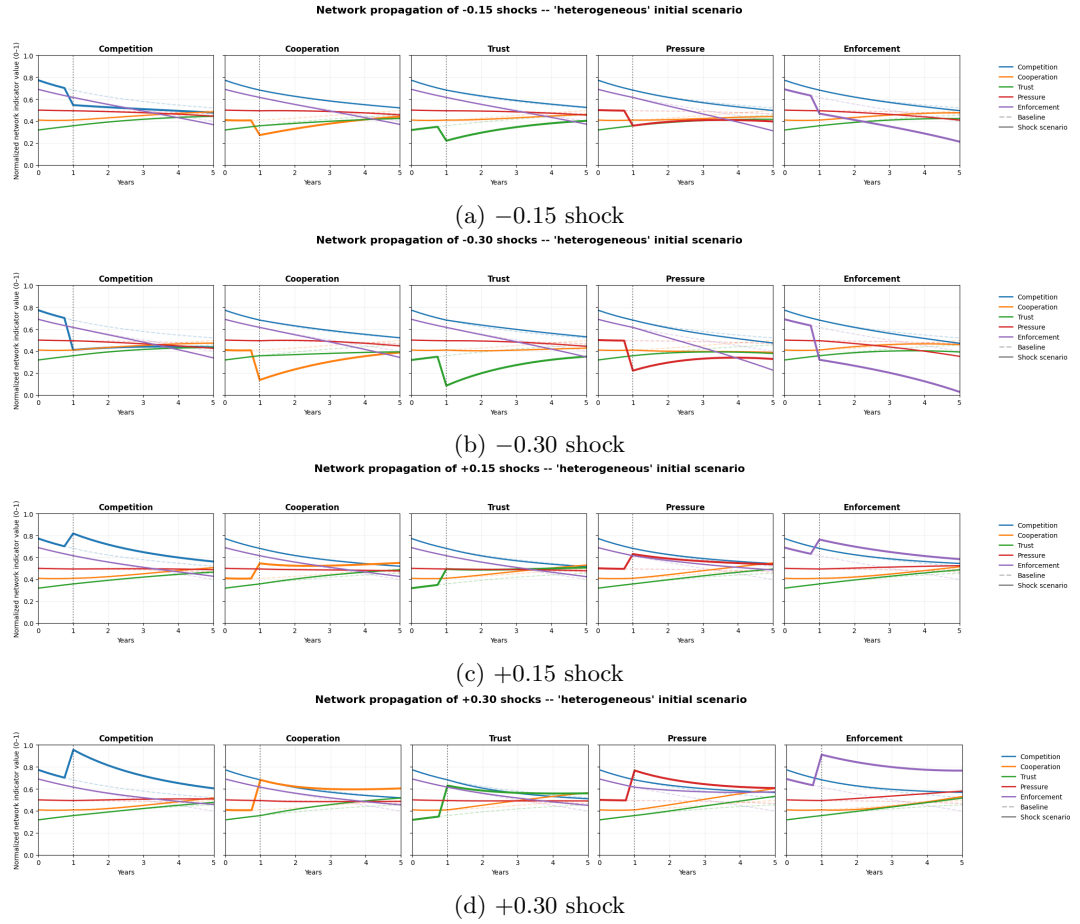


Fig. 7: Time-series response of network indicator values (0–1) over a 5-year period following single-parameter shock interventions for heterogeneous initial scenario. Dashed lines indicate unshocked baseline trajectories; bold solid lines capture immediate parameter displacement at Year 1 and subsequent network propagation. Nullpoint and balance scenario are shown in Figures 9 and 10.

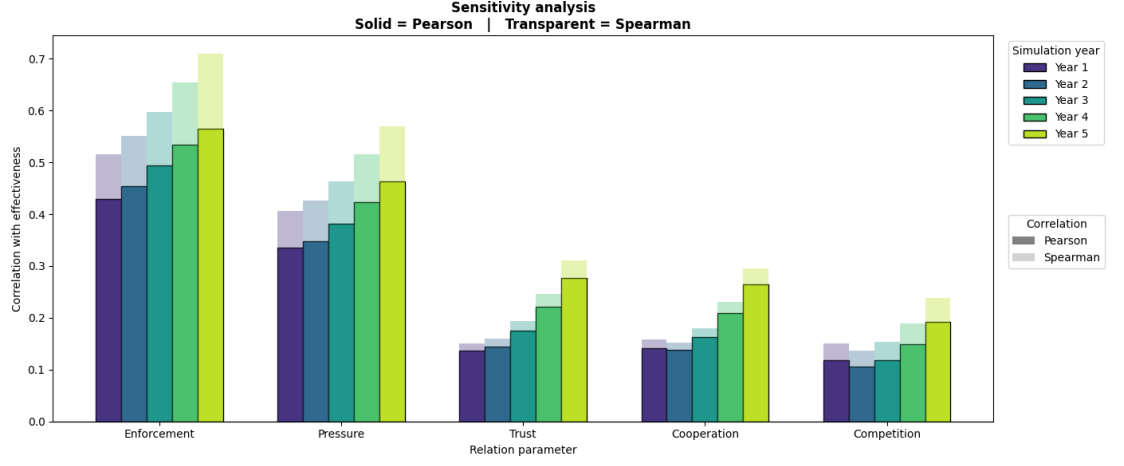


Fig. 8: Parameter sensitivity across shock combinations in the heterogeneous baseline. Institutional mechanisms dominate effectiveness outcomes, with enforcement and pressure showing the strongest associations ($r = 0.429$, $\rho = 0.516$; $r = 0.335$, $\rho = 0.406$), while relational mechanisms exhibit weaker overall influence, as reflected in smaller correlations for cooperation, trust, and competition.

Enforcement decay:

$$E_{t+1} = E_t \times (1 - \delta_E) + \sum \text{Inputs}$$

where $\delta_E = 0.015$. Systemic parameters also experience homeostatic relaxation toward baseline equilibrium states (θ), formulated as:

$$\Delta P_i = \gamma(P_{i,\text{baseline}} - P_i) + \sum_j w_{ji} P_j$$

where γ represents the relaxation coefficient governing network recovery speed, and w_{ji} represents the weight of influence from parameter j to parameter i .

Pressure is treated as a conditional variable whose operational contribution to PPC effectiveness (P) is modeled as a non-monotonic function of its intensity. Rather than acting strictly as a reinforcing or constraining driver, pressure follows an inverted-U response curve: effectiveness drops to zero below an operational floor ($\theta_{\text{floor}} = 0.2$), while values above the threshold are evaluated via a Gaussian response centered at an optimal value ($\mu = 0.5$, $\sigma = 0.28$):

$$P(p) = \begin{cases} 0, & \text{if } p \leq \theta_{\text{floor}} \\ \exp\left(-\frac{1}{2} \left(\frac{p-\mu}{\sigma}\right)^2\right), & \text{if } p > \theta_{\text{floor}} \end{cases}$$

where $p \in [0, 1]$ represents the aggregate network pressure at time step t . Below 0.2, actors lack sufficient urgency to collaborate; beyond 0.5, escalating pressure produces counterproductive friction and trust degradation.

Table 4: Overview of implemented ABM rules and parameter origins.

Rule / Parameter	Implementation	Origin
Skill-to-parameter influence	Actor skills gradually influence relational parameters (e.g., cooperation and trust).	DE (direction), assumption (rate)
Parameter-to-skill influence	Relational parameters influence actor skills over time.	DE (direction), assumption (rate)
Parameter feedback	Trust, cooperation, competition, pressure, and enforcement influence one another.	DE (direction), assumption (speed)
Minimum pressure requirement	Effective cooperation requires pressure > 0.2.	DE
Pressure optimum	Cooperation is highest at moderate pressure levels.	DE, assumption (magnitude)
Public pressure advantage	Public actors apply pressure more effectively.	DE, assumption (magnitude)
Private competition advantage	Private actors exhibit stronger competitive behaviour.	DE, assumption (magnitude)
Institutional requirements	At least one public, private, authoritative, and operational actor is required.	DE
Risk mentality requirement	Risk mentality must exceed punishment mentality.	DE
Resource requirements	Economic, cultural, and social capital, DE and urgency are required.	DE
Relationship dissolution	Very weak network ties are removed below a threshold.	Assumption
Relationship relaxation	Relational parameters gradually return to equilibrium.	Assumption
Skill relaxation	Actor skills gradually return to baseline values.	Assumption
Enforcement decay	Enforcement decreases each simulation step unless renewed.	DE, assumption (rate)
Reputation penalty	Higher average reputation reduces PPC effectiveness.	DE, assumption (weight)

Table 5: Simulation parameter values used in the ABM.

Parameter	Value	Description
Skill $\rightarrow$ parameter rate	0.030	Skill-to-parameter propagation speed
Parameter $\rightarrow$ skill rate	0.035	Parameter-to-skill propagation speed
Parameter feedback rate	0.030	Parameter interaction speed
Enforcement decay	0.985	Multiplicative decay per simulation step
Reputation penalty weight	0.15	Penalty applied to effectiveness
Public enforcement boost	1.15	Public actor enforcement multiplier
Public pressure boost	1.15	Public actor pressure multiplier
Private competition boost	1.15	Private actor competition multiplier
Pressure optimum	0.50	Centre of inverted-U response
Pressure width	0.28	Width of effective pressure region
Pressure floor	0.20	Minimum pressure threshold
Tie dissolution threshold	0.15	Minimum network tie strength
Indicator relaxation	0.09	Mean reversion of relation indicators
Skill relaxation	0.05	Mean reversion of actor skills

Table 6: Top 5 intervention strategies from the exhaustive combinatorial shock analysis (heterogeneous scenario).

Rank	Shock pattern	PPC effectiveness	effect- Δ Effectiveness
1	Competition -0.6 , Cooperation $+0.6$, Trust $+0.7$, Pressure $+0.5$, Enforcement -0.5	0.568	0.246
2	Competition -0.8 , Cooperation $+0.5$, Trust $+0.7$, Pressure $+0.5$, Enforcement -0.4	0.568	0.245
3	Competition -0.8 , Cooperation $+0.6$, Trust $+0.6$, Pressure $+0.5$, Enforcement -0.4	0.568	0.245
4	Competition -0.7 , Cooperation $+0.6$, Trust $+0.7$, Pressure $+0.4$, Enforcement -0.4	0.567	0.245
5	Competition -0.8 , Cooperation $+0.6$, Trust $+0.7$, Pressure $+0.5$, Enforcement -0.4	0.565	0.243

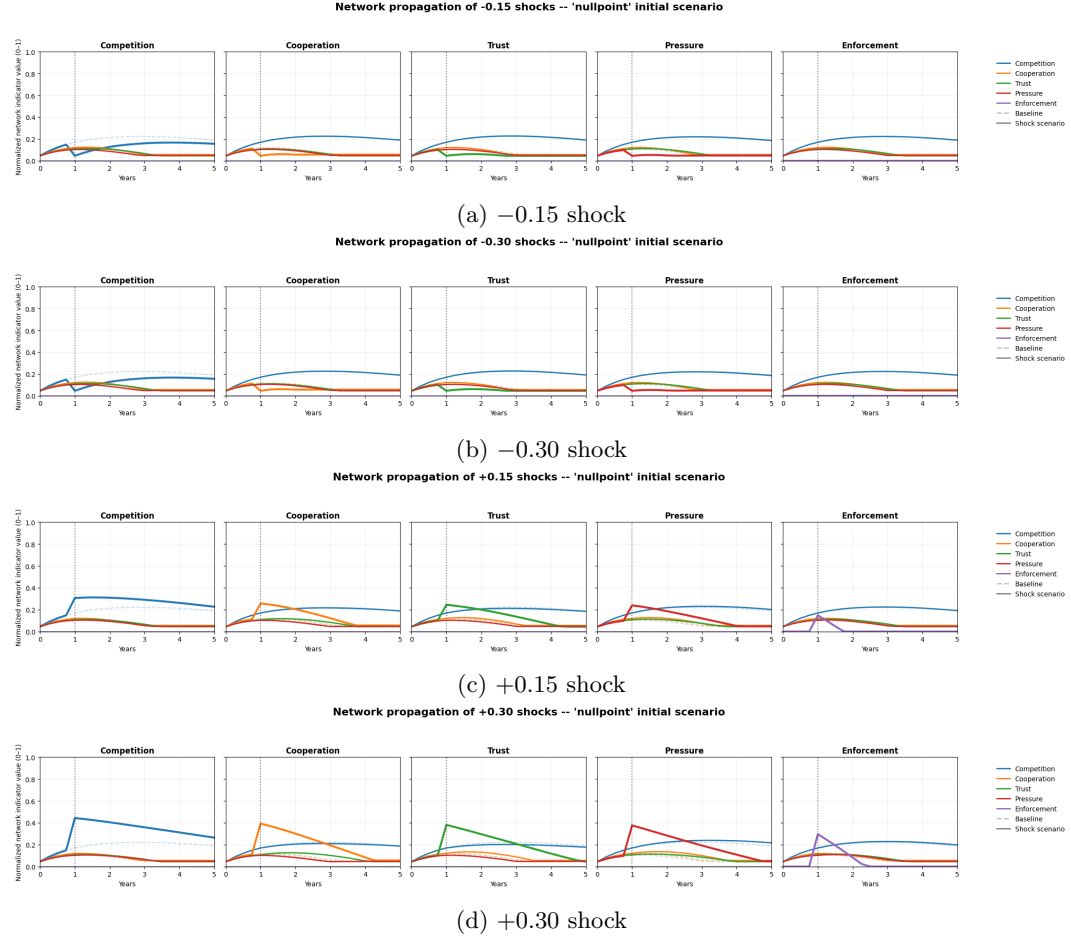


Fig. 9: Time-series response of network indicator values (0–1) over a 5-year period following single-parameter shock interventions for the nullpoint initial scenario. Dashed lines indicate unshocked baseline trajectories; bold solid lines indicate the immediate parameter displacement at Year 1 and subsequent network propagation.

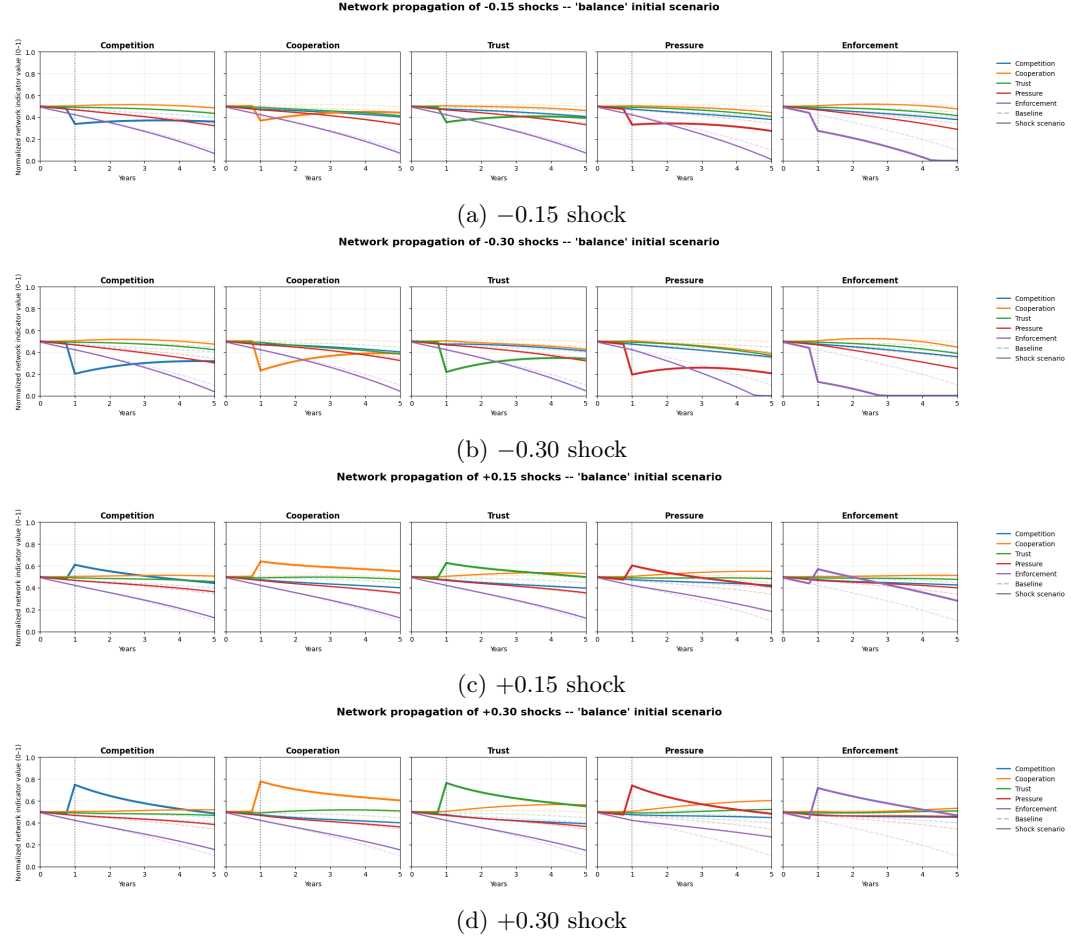


Fig. 10: Time-series response of network indicator values (0–1) over a 5-year period following single-parameter shock interventions for the balance initial scenario. Dashed lines indicate unshocked baseline trajectories; bold solid lines indicate the immediate parameter displacement at Year 1 and subsequent network propagation.

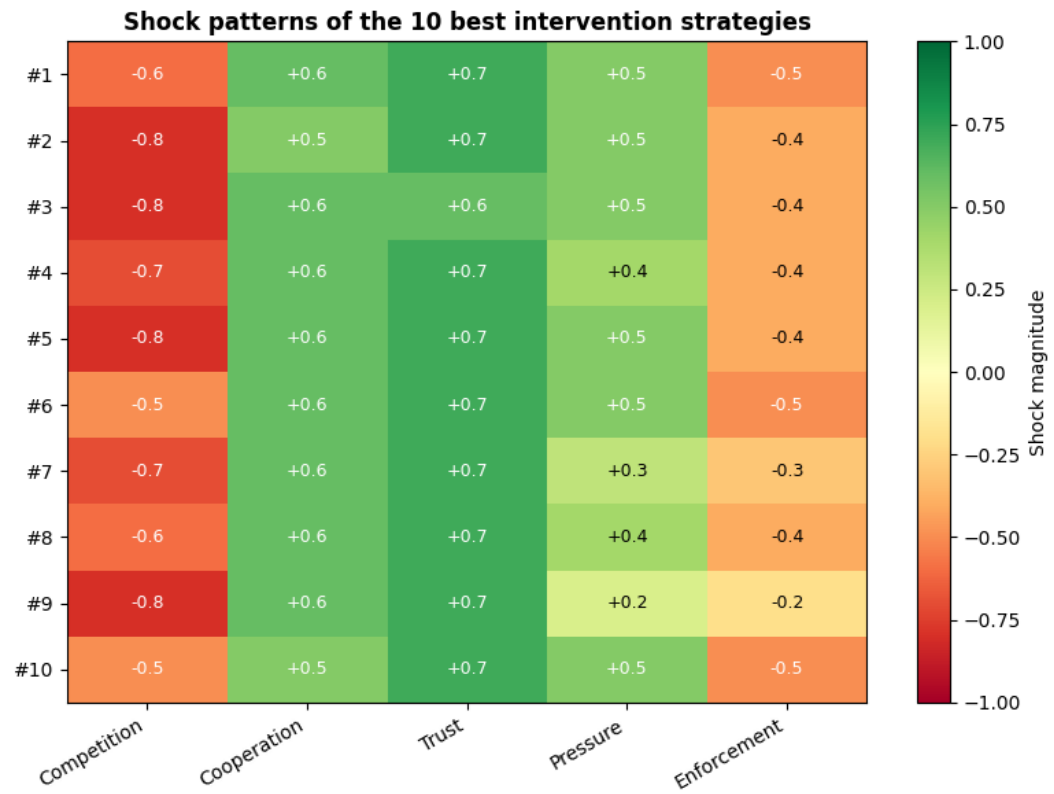


Fig. 11: Top 10 best intervention strategies in regard to effectiveness for heterogeneous scenario with multi-parameter analysis.

The PPC effectiveness score is calculated as

$$\text{PPC}_{\text{eff}} = \text{clip} (S \cdot P \cdot (0.25T + 0.25C + 0.20R + 0.15M + 0.15Q) \cdot (0.5 + 0.5F) - H, 0, 1), \quad (1)$$

where

- S denotes the structural validity gate;
- P denotes the pressure sweet-spot factor;
- T denotes the trust trend gate;
- C denotes the cooperation trend gate;
- R denotes the cooperation–competition gate, requiring cooperation to exceed competition;
- M denotes the mentality gate;
- Q denotes the resource gate;
- F denotes the responsibility term; and
- H denotes the reputation penalty.

The effectiveness score is not a simple average of the five relational indicators. Instead, it represents the degree to which the overall system satisfies a set of institutional, relational, behavioural, and resource conditions considered necessary for successful PPC.

The structural validity gate is binary and ensures that the network contains at least one public actor, one private actor, an actor with decision-making authority, and an actor operating at the operational level. If any of these conditions are not satisfied, PPC effectiveness is reduced to zero. The pressure component follows an inverted-U relationship, reflecting the assumption that both insufficient and excessive pressure reduce collaborative performance. Trust and cooperation are required to remain positive and exhibit an increasing trend over time, while cooperation must exceed competition to favour collaborative rather than adversarial interactions.

The mentality component combines the average levels of risk mentality and punishment mentality, rewarding willingness to cooperate while penalizing excessive aversion to risk. The resource component requires sufficient economic, cultural, and social capital together with an adequate sense of urgency. Formal and informal responsibility jointly increase effectiveness through a multiplicative scaling factor, whereas reputation contributes as a penalty term under the assumption that highly reputation-sensitive actors become increasingly risk-averse over time. After all components have been combined, the resulting effectiveness score is clipped to the interval $[0, 1]$, ensuring that the model output remains a normalized measure of overall PPC effectiveness.